

MPC-MAP Assignment No. 5 - Report

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Task 1,2

I implemented the A* (A-star) algorithm to find an optimal, collision-free path in the occupancy grid using an 8-connected neighborhood and Euclidean distance heuristic. The algorithm dynamically retrieves the start (mu from EKF) and goal positions, ensuring full universality across all indoor and outdoor maps. To maintain a safe clearance of at least 0.2 m from obstacles, I applied a map inflation technique. Instead of relying on the Image Processing Toolbox, I used 2D convolution (conv2) with a square kernel to efficiently expand all obstacles by approximately 0.25 m before running the A* search.

Task 3

To eliminate the sharp, grid-aligned turns of the raw A* output, I applied an iterative smoother. The smoothing behavior is driven by two key parameters. The `weight_smooth` parameter controls the tendency to straighten the path; if set too high, the path contracts and dangerously cuts corners into the obstacle clearance zones. The `weight_data` parameter controls the anchoring to the original A* waypoints; if set too high, the algorithm fails to smooth out sharp turns. By tuning the parameters to `weight_data = 0.5` and `weight_smooth = 0.1`, I achieved the perfect balance, resulting in a continuously smooth trajectory that safely respects all obstacle boundaries.

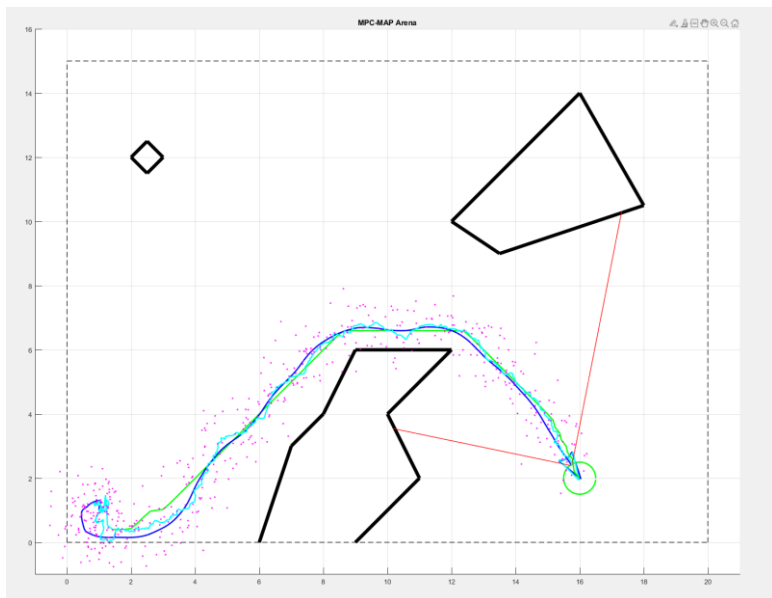


Figure 1: Path planning and obstacle avoidance

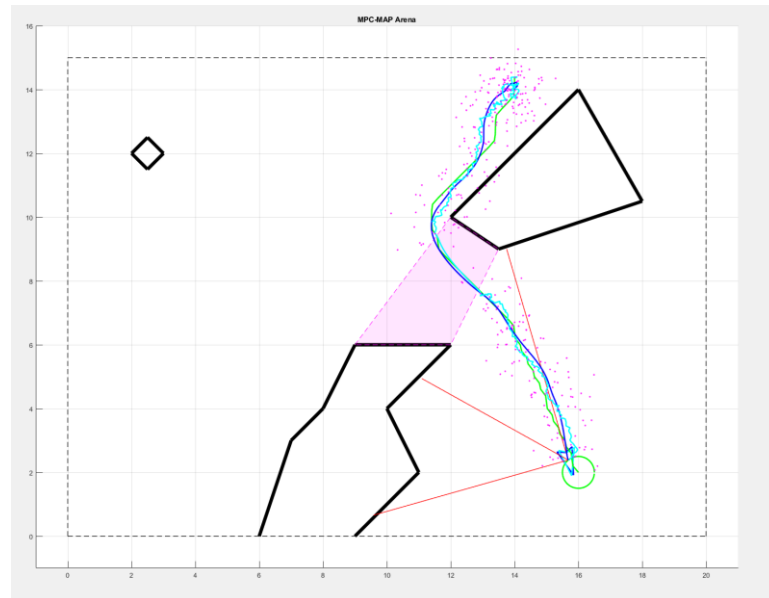


Figure 2: navigation with GNSS outage